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How a Project at IIT Delhi Gave a Fillip to India's Defence Tech

I WAS NOT A TYPICAL IIT STUDENT. After joining the National Defence Academy in Khadakwasla, Maharashtra, in 1961, I spent seven years in service and engineering training. Following a brief stint on warships, the Navy sent me to IIT Delhi in July 1969 for a two-year MTech programme. The IIT, with its impressive labs and libraries, offered a different world from the Navy and a new opportunity.

At IIT, Professor P.V. Indiresan (who later served as the director of IIT Madras) helped me switch to a PhD programme within my approved two-year Navy deputation. As a PhD scholar, I worked on a rather abstract theory of non-linear filtering, utilising tools from stochastic calculus and diffusion models. This work caught the attention of researchers at Stanford University and UC Berkeley, eventually leading to my faculty position at Stanford two decades later.

After my two years at IIT Delhi, I returned to the Navy in July 1971. Unexpectedly, I was back at the institute in early 1972 to lead a sonar redesign project, far removed from my PhD research. This project followed the Indo-Pak war of December 1971, where the Navy lost INS Khukri, a frontline anti-submarine frigate, along with 194 lives. A Pakistani submarine, PNS Hangor, had successfully torpedoed Khukri. The British-designed 170B sonar on Khukri had apparently failed to detect Hangor. The Navy tasked me with developing an improved design for the 170B, which I naturally chose to undertake at IIT Delhi, leading a small team of DRDO scientists and PhD students.

The 18-month project yielded a significantly enhanced transmitter and receiver system, leveraging advanced signal processing technology. After extensive sea trials, the Navy approved the equipment for production in 1973. A defence-related production unit manufactured several systems for deployment in the Indian fleet. This achievement—conceived at IIT Delhi—marked the first successful development of a complex electronic system in independent India, inspiring later successes in Indian defence technology.

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My IIT days left me with many pleasant memories. As a naval officer, I resided at the faculty house instead of student hostels, and was grateful for the numerous comforts it provided. I also forged lasting friendships that have endured for decades.

Prof. Indiresan and IIT Delhi gave me, a naval sailor, a chance at an R&D career. After 20 years in India, I moved to Stanford University, and with that came many opportunities. At Stanford, I invented MIMO (Multiple-Input Multiple-Output) wireless, which became the core technology for today's 4G/5G mobile and WiFi networks used worldwide. I am indeed most grateful for the formative years at IIT Delhi. ■

The author, a globally recognised pioneer of modern wireless communications technology, is an Emeritus Professor at Stanford University, USA. He is a 1973 PhD graduate from IIT Delhi